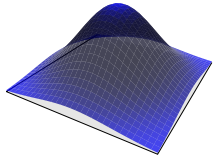
TECHNISCHE UNIVERSITÄT WIEN



Lecture 2

Statistics and Probability Theory

Combinatorics and Probability Spaces

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Example: Counting

The brute force approach

What is the *probability* of getting *exactly 2 heads* out of *4 tosses* of a fair coin?

SAMPLE SPACE: Listing all 16 *possible outcomes*

$$\Omega = \{HHHH, HHHT, HHTH, HHTT, HTHH, HTHT, HTTH, HTTT,$$
$$THHH, THHT, THTH, THTT, TTHH, TTHT, TTTH, TTTT\}$$

EVENT: There are *6 outcomes* with *"exactly 2 heads"*

$$A = \{HHTT, HTHT, HTTH, THHT, THTH, TTHH\}$$

PROBABILITY: With all outcomes being equally likely; we get

$$P(A) = \frac{|A|}{|\Omega|} = \frac{\text{6 outcomes with 2 heads}}{\text{16 possible outcomes}} = \frac{6}{16} = 37.5\%.$$

What about 3 heads in 10 tosses? Or 7 heads in 100 tosses?

SET

A *set* is a *collection of distinct elements*.

Notation and operations for sets A and B in a "universe" Ω;

$x \in A$	element x is in A
$A \subseteq B$	A is a subset of B [2]
$\emptyset$	the empty set (set without elements)
$A \cap B = \{x : x \in A \text{ and } x \in B\}$	intersection
$A \cup B = \{x : x \in A \text{ or } x \in B\}$	union
$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$	difference
$A \triangle B = (A \setminus B) \cup (B \setminus A)$	symmetric difference
$A^c = \Omega \setminus A$	complement (set of elements *not* in A)
$\lvert A \rvert$	number of elements in A (size of A)

[2] also written as $A \subset B$, which may also be used for a "proper subset" $A \subsetneq B$.

Some properties of sets A, B in a "universe" Ω;

Commutativity

$$A \cap B = B \cap A \qquad A \cup B = B \cup A$$

Associativity

$$(A \cap B) \cap C = A \cap (B \cap C) \qquad (A \cup B) \cup C = A \cup (B \cup C)$$

Distributivity

$$(A \cup B) \cap C = (A \cap C) \cup (B \cap C) \qquad (A \cap B) \cup C = (A \cup C) \cap (B \cup C)$$

Laws of Complement

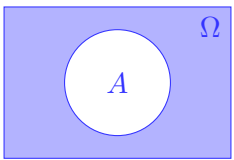
$$A \cup A^c = \Omega \qquad A \cap A^c = \emptyset \qquad (A^c)^c = A \qquad \Omega^c = \emptyset$$

De Morgan's laws

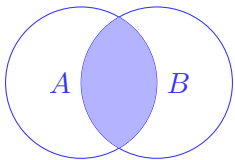
$$(A \cup B)^c = A^c \cap B^c \qquad (A \cap B)^c = A^c \cup B^c$$

Venn diagrams provide an easy way of visualizing set operations;

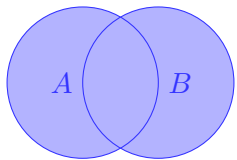
A^c



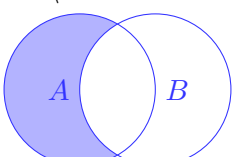
$A \cap B$



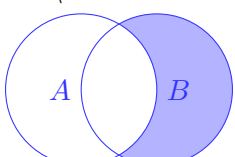
$A \cup B$



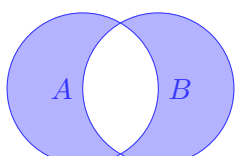
$A \setminus B = A \cap B^c$



$B \setminus A = A^c \cap B$



$A \triangle B$



The *Cartesian product* of two sets A and B is the *set of all ordered pairs*

$$A \times B = \{(a, b) : a \in A \text{ and } b \in B\}$$

For example for $A = \{1, 2, 3\}$ and $B = \{2, 3, 4, 5\}$ we get

$$A \times B = \{(1,2), (1,3), (1,4), (1,5), (2,2), (2,3), (2,4), (2,5), (3,2), (3,3), (3,4), (3,5)\}$$

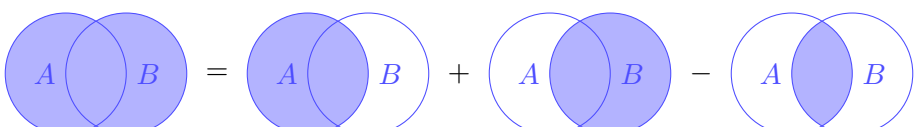
which is better displayed via a table

$A \times B$		B			
		2	3	4	5
	1	(1, 2)	(1, 3)	(1, 4)	(1, 5)
A	2	(2, 2)	(2, 3)	(2, 4)	(2, 5)
	3	(3, 2)	(3, 3)	(3, 4)	(3, 5)

Inclusion-Exclusion Principle

The size of the union of two finite sets A, B is

$$|A \cup B| = |A| + |B| - |A \cap B|.$$



The same is applicable for 3 finite sets

$$|A \cup B \cup C| = |A| + |B| + |C| - |A \cap B| - |A \cap C| - |B \cap C| + |A \cap B \cap C|$$

or for any finite union of finite sets

$$\left|\bigcup_{i \in I} A_i\right| = \sum_{\emptyset \neq J \subseteq I} (-1)^{|J|+1} \left|\bigcap_{j \in J} A_j\right|$$

using $I = \{1, 2, \ldots, n\}$ and sum over all non-empty index subsets $J \subseteq I$.

A band consists of singers and guitar players;

S 7 singers

G 4 guitar players

$S \cap G$ 2 both sing and play guitar

How many people are in the band?

$$\text{Size of the Band} = |S \cup G| = |S| + |G| - |S \cap G| = 7 + 4 - 2 = 9.$$

How many numbers from 1 to 1 000 are neither divisible by 2 nor 3? [3]
Therefore, let

$$A_2 = \{2, 4, 6, \ldots, 998, 1\,000\},$$
$$A_3 = \{3, 6, 9, \ldots, 996, 999\}$$

be the set of all 2 and 3 divisible numbers, respectively.
Their intersection is easy to describe, its the set of all $6 = 2 \times 3$ divisible numbers

$$A_6 = A_2 \cap A_3 = \{6, 12, 18, \ldots, 990, 996\}.$$

Their size is easily given as[4] $|A_2| = 500$, $|A_3| = 333$ and $|A_6| = 166$.
We are interested in the size of $(A_2 \cup A_3)^c$, which we can compute as

$$\begin{aligned} |(A_2 \cup A_3)^c| &= |\Omega| - |A_2 \cup A_3| \\ &= |\Omega| - (|A_2| + |A_3| - |A_2 \cap A_3|) = 1\,000 - (500 + 333 - 166) = 333 \end{aligned}$$

[3] Those are the numbers $1, 5, 7, 11, 13, 17, 19, 23, 25, 29, \ldots 983, 985, 989, 991, 995, 997$
[4] Divide max element in the set by its index $|A_k| = \max(A_k)/k$.

Factorial

For $n \in \mathbb{N}_0$, the factorial is the product of all number up to n

$$n! = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1$$

For 0 we define $0! = 1$.

$0! = 1$ $\quad$ $1! = 1$

$2! = 2 \cdot 1 = 2$ $\quad$ $3! = 3 \cdot 2 \cdot 1 = 6$

$4! = 4 \cdot 3! = 24$ $\quad$ $9! = 362\,880$

Binomial Coefficient

The binomial coefficient for $n, k \in \mathbb{N}_0$ with $0 \leq k \leq n$ is

$$\binom{n}{k} = \frac{n!}{k!(n-k)!}$$

Spoken as "n choose k".

$$\binom{7}{3} = \frac{7!}{3!\,4!} = 35$$

Product Rule

If there are *n ways to perform action 1* followed by *m ways to perform action 2*, then there are *$n \cdot m$ ways to perform action 1 followed by action 2*.

For example consider 8 participants in a tournament. How many ways are there to awarded gold, silver and bronze medals?

$$8 \cdot 7 \cdot 6 = 336$$

Permutation
without repetition

A *permutation* is a *rearrangement of distinct elements*.
Ordered!

The sequence $(1, 2, 3)$ can be rearranged in the follwoing ways

$$(1, 2, 3), (1, 3, 2), (2, 1, 3), (2, 3, 1), (3, 1, 2), (3, 2, 1)$$

By the product rule, we get

$$n! = n(n-1)(n-2)\cdots 3 \cdot 2 \cdot 1$$

permutations of n elements.

A poker deck consists of 52 distinct cards.

How many different shuffles of a poker deck are there?

$$52! \approx 8 \cdot 10^{67}$$

or exactly

$$52! = 80\,658\,175\,170\,943\,878\,571\,660\,636\,856\,403\,766\,975\,289\,505\,440\,883\,277\,824\,000\,000\,000\,000$$

VARIATION
without repetition

Variations without repetition are *ordered selections of k different elements out of n.*
Ordered!

Out of $\{1, 2, 3\}$ all size 2 tuples are $(1, 2), (2, 1), (1, 3), (3, 1), (2, 3), (3, 2)$.

By the product rule we get

$$n(n-1)\cdots(n-k+2)(n-k+1) = \frac{n!}{(n-k)!} = k!\binom{n}{k}$$

many variations without repetition.

In a race of 8 runners, how many variations of gold, silver and bronze medalists are there?

$$\frac{8!}{(8-3)!} = 8 \cdot 7 \cdot 6 = 336$$

Combination
without repetition

A *combination without repetition* is a *selection of k elements out of n elements*

or

a *subset of size k from a set of size n.*

Unordered!

The set $\{1, 2, 3\}$ has 3 subsets of size 2, namely $\{1, 2\}, \{1, 3\}, \{2, 3\}$.

By the product rule we get $n(n-1)\cdots(n-k+2)(n-k+1)$ ways of selecting elements in order. Since the order is irrelevant, we divide[5] by $k!$ permutations of the selected elements to get the number of combinations

$$\frac{n(n-1)\cdots(n-k+2)(n-k+1)}{k!} = \frac{n!}{k!(n-k)!} = \binom{n}{k}$$

[5] Divisibility is ensured, always an integer!

In poker, a hand consists of 5 cards out of a set of 52 cards.

How many different hands are there?

$$\binom{52}{5} = \frac{52!}{5!(52-5)!} = \frac{52 \cdot 51 \cdot 50 \cdot 49 \cdot 48}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1} = 26 \cdot 17 \cdot 10 \cdot 49 \cdot 12 = 2\,598\,960$$

Permutation
with repetition

A *permutation with repetitions* is a *rearrangement of n elements consisting of k distinct elements*

Ordered!

All 30 permutations of $n = 5$ elements $(1, 1, 2, 2, 3)$ out of $k = 3$ different elements.

11223	11232	11322	12123	12132	12213	12231	12312	12321	13122
13212	13221	21123	21132	21213	21231	21312	21321	22113	22131
22311	23112	23121	23211	31122	31212	31221	32112	32121	32211

There are $n!$ permutations of n elements. Each n_i times repetition of the same element is overcounted by $n_i!$. With $n_1 + n_2 + \ldots + n_k = n$ we get the count;

$$\frac{n!}{n_1! n_2! \cdots n_k!}$$

A poker deck has 52 cards and 4 suits ($\heartsuit, \spadesuit, \diamondsuit, \clubsuit$).

Treating all the card values equally and only considering the suit ($\mathrm{Q}\heartsuit = \mathrm{K}\heartsuit$ but $\mathrm{Q}\heartsuit \neq \mathrm{Q}\spadesuit$), how many distinguishable shuffles are there?

We have $n = 52$ cards with $k = 4$ suits, each of size $n_\heartsuit = n_\spadesuit = n_\diamondsuit = n_\clubsuit = 13$, which gives

$$\frac{n!}{n_\heartsuit!\, n_\spadesuit!\, n_\diamondsuit!\, n_\clubsuit!} = \frac{52!}{13!\, 13!\, 13!\, 13!} = 53\,644\,737\,765\,488\,792\,839\,237\,440\,000$$

Variation
with repetition

Variations with repetition are *ordered repeated selections of k elements out of n.*
Ordered!

All 9 variations of $\{1, 2, 3\}$ of size $k = 2$ are

$$(1,1), (1,2), (1,3), (2,1), (2,2), (2,3), (3,1), (3,2), (3,3)$$

By the product rule we get

$$n^k$$

variations with replacement (k might be bigger than n).

A 64 bit unsigned integer is a sequence of 64 zeros or ones.

How many numbers can be represented?

$$2^{64} = 18\,446\,744\,073\,709\,551\,616$$

Combination
with repetition

Combinations with repetition are *repeated selections of k elements out of n.*

or

size k mutli-sets of elements from a set of size n.

Unordered!

Choosing $k = 3$ out of $\{1, 2, 3, 4, 5\}$ ($n = 5$) with repetition gives 35 combinations

$\{1,1,1\}$	$\{1,2,2\}$	$\{1,3,4\}$	$\{2,2,2\}$	$\{2,3,4\}$	$\{3,3,3\}$	$\{3,5,5\}$
$\{1,1,2\}$	$\{1,2,3\}$	$\{1,3,5\}$	$\{2,2,3\}$	$\{2,3,5\}$	$\{3,3,4\}$	$\{4,4,4\}$
$\{1,1,3\}$	$\{1,2,4\}$	$\{1,4,4\}$	$\{2,2,4\}$	$\{2,4,4\}$	$\{3,3,5\}$	$\{4,4,5\}$
$\{1,1,4\}$	$\{1,2,5\}$	$\{1,4,5\}$	$\{2,2,5\}$	$\{2,4,5\}$	$\{3,4,4\}$	$\{4,5,5\}$
$\{1,1,5\}$	$\{1,3,3\}$	$\{1,5,5\}$	$\{2,3,3\}$	$\{2,5,5\}$	$\{3,4,5\}$	$\{5,5,5\}$

How many combinations with replacement (or multi-sets of size k) are there?

Consider a multi-set of size $k = 6$ out of $n = 5$ elements $\{1, 2, 3, 4, 5\}$, for example

$$\{1, 2, 2, 2, 4, 4\}.$$

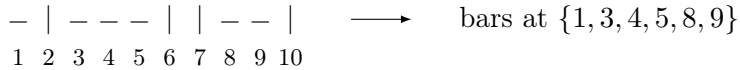
We can encoded this using bars and stripes as

$$\underset{1\times 1}{-}\ |\ \underbrace{-\,-\,-}_{3\times 2}\ |\ |\ \underbrace{-\,-}_{2\times 4}\ |\ \underset{0\times 3}{}\ \underset{0\times 5}{}$$

which has $k = 6$ bars and $n - 1 = 4$ stripes at $n - 1 + k = 10$ positions.

The bars and stripes encoding is a one-to-one relation.

The one-to-one relation allows to count the bars and stripes encoding instead.



Every encoding corresponds to a k set of bar positions out of $n - 1 + k$ options.

Those are the *combinations without replacement of k elements out of $n - 1 + k$*

$$\binom{n-1+k}{k}$$

which is the number of *combinations of k elements out of n with replacement*.

When playing black-jack at a casino, it is common that multiple decks are used.

Assuming 5 decks are used, how many 5 card draws are possible?

$$\binom{52-1+5}{5} = \frac{56!}{5!\,51!} = \frac{56 \cdot 55 \cdot 54 \cdot 53 \cdot 52}{5 \cdot 4 \cdot 3 \cdot 2} = 28 \cdot 11 \cdot 18 \cdot 53 \cdot 13 = 3\,819\,816$$

Combinatorics

	Ordered		Unordered
	Permutations	Variations	Combinations
without repetition	$n!$	$\frac{n!}{(n-k)!}$	$\binom{n}{k}$
with repetition	$\frac{n!}{n_1!n_2!\cdots n_k!}$	n^k	$\binom{n-1+k}{k}$

The formal mathematical description of probability is based on the concept of a

Probability Space

A *probability space* is a tuple $(\Omega, \mathfrak{S}, P)$ consisting of a *sample space* Ω, a corresponding σ-algebra $\mathfrak{S}$ and a *probability measure* $P : \mathfrak{S} \to [0, 1]$.

Sample Space: The sample space contains all *elementary events* $\omega \in \Omega$.

σ-Algebra: Till now we only considering finite sample spaces, which implies that $\mathfrak{S} = 2^{\Omega}$ is the power-set (the set of all subsets).[-1]

Probability Measure: A *probability measure* assignes every possible event $A \in \mathfrak{S}$ a probability (a numerical value in the range $[0, 1]$).

[-1] Not important for you! Anyway, I want to save guard you against measuring unmeasurable sets, which might annihilate our universe and replace it with something even more complicated. A big problem in philosophy of how often this has occured.

Sample Space: collection of all possible (elementary) events.

Probability: sum up all elementary event probabilities or the integral over all events with respect to the probability measure (see the future).

For example consider we are playing D&D and fight against a goblin with 4 health points. We ourself enflich one 6-sided die of damage. If we role a 1 or 2 the gobline is fine and attacks us next turn, if we role a 3 the goblin in unconchious, any other role kills the goblin.

This can be described as

Ω: {goblin fine, goblin unconchious, goblin dead}

P: $P(\{\text{goblin fine}\}) = \frac{1}{3}$, $P(\{\text{goblin unconchious}\}) = \frac{1}{6}$, $P(\{\text{goblin dead}\}) = \frac{1}{2}$

$$P(\text{goblin alive}) = P(\{\text{goblin is fine or unconchious}\}) = \frac{1}{3} + \frac{1}{6} = \frac{1}{2}$$